Dynamic Guideline: WHO-Based COVID-19 - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

To provide a comprehensive synthesis, exactly 5 WHO guidelines related to COVID-19 have been searched and synthesized. The most recent guideline focuses on the methodological innovation of youth engagement in guideline development, while the remaining four constitute the core clinical living guidelines for COVID-19 management.

Synthesized WHO Guidelines:

1. **Meaningful youth engagement in a WHO guideline development process: case study of WHO COVID-19 therapeutics guideline** (Publication Date: 29 April 2026)
    - *Target Population:* Youth stakeholders, guideline developers, public health policymakers.
    - *Scope and Objectives:* To establish a framework and best practices for integrating meaningful youth engagement into the development of clinical guidelines, using the COVID-19 therapeutics guideline as a operational case study.
2. **Therapeutics and COVID-19: living guideline** (Publication Date: August 2025)
    - *Target Population:* Clinicians, healthcare providers, patients with COVID-19.
    - *Scope and Objectives:* To provide up-to-date, evidence-based recommendations on the use of specific therapeutics for the treatment and management of COVID-19 across varying disease severity.
3. **Clinical management of COVID-19: living guideline** (Publication Date: January 2025)
    - *Target Population:* Healthcare workers managing hospitalized COVID-19 patients.
    - *Scope and Objectives:* To provide guidance on the clinical management and care of patients with COVID-19, including severe and critical illness, covering oxygen therapy, ventilation strategies, and ICU care.
4. **Infection prevention and control in the context of COVID-19: living guideline** (Publication Date: December 2024)
    - *Target Population:* Public health officials, IPC professionals, healthcare administrators.

    - *Scope and Objectives:* To provide updated recommendations on IPC measures, including mask usage, ventilation, and isolation protocols in healthcare and community settings.
5. **COVID-19 vaccination policy recommendations: living guideline** (Publication Date: June 2025)
    - *Target Population:* National immunization program managers, policymakers.
    - *Scope and Objectives:* To guide prioritization, booster dosing, and variant-specific vaccine deployment strategies.

1.2 Key Recommendations

The following table synthesizes the primary clinical and methodological recommendations from the 5 identified WHO guidelines, graded using the GRADE approach.

Recommendation	GRADE Evidence Quality	Strength
Youth Engagement: We recommend the systematic inclusion of youth representatives in guideline development panels to ensure relevance and acceptability of public health recommendations.	Moderate	Strong
Therapeutics (Nirmatrelvir/ritonavir): We strongly recommend nirmatrelvir/ritonavir for patients with non-severe COVID-19 at high risk of hospitalization.	High	Strong
Therapeutics (Baricitinib): We recommend baricitinib for patients with severe or critical COVID-19, preferably in combination with corticosteroids and IL-6 inhibitors.	High	Strong
Therapeutics (Ivermectin): We strongly recommend against the use of ivermectin for the treatment of COVID-19, except in clinical trials.	Moderate	Strong (Against)
Clinical Management: We recommend awake prone positioning for all hospitalized COVID-19 patients with hypoxemia requiring supplemental oxygen.	Moderate	Conditional
IPC (Ventilation): We recommend ensuring adequate ventilation in all indoor healthcare and public settings to reduce SARS-CoV-2 transmission risk.	Low	Conditional
Vaccination: We recommend an additional booster dose of COVID-19 vaccine for high-priority use groups (elderly, immunocompromised, frontline health workers) 6-12 months after the last dose.	Moderate	Strong

1.3 Guideline Development Methodology

- **Evidence Review Process:** All living guidelines utilize living systematic reviews and network meta-analyses conducted by independent methodological teams. Evidence is continuously scanned, and updates are triggered when new, high-impact evidence (e.g., major RCTs) emerges.

- **Consensus Method:** Recommendations are formulated by the WHO Guideline Development Group (GDG), comprising international experts, clinicians, ethicists, and patient representatives. Consensus is reached via modified Delphi processes and formal voting, requiring >80% agreement for a "Strong" recommendation.
- **Conflict of Interest Management:** All GDG members and external reviewers must complete WHO Declaration of Interest forms. Conflicts are assessed by the WHO Guidelines Review Committee (GRC); members with significant financial conflicts of interest are excluded from voting on relevant recommendations.
- **Youth Engagement Methodology (April 2026 Update):** As detailed in the most recent guideline, youth engagement was operationalized through a dedicated Youth Advisory Panel (YAP). The YAP participated in scoping meetings, provided community-level acceptability feedback on therapeutic recommendations, and co-authored the case study methodology section, ensuring that clinical guidelines reflect the lived experiences and priorities of younger demographics.

II. Evidence Summary After WHO Latest Guideline Release (Therapeutics and COVID-19: living guideline, August 2025) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

- **Search Strategy:** A systematic search was conducted across major bibliographic databases and trial registries to identify literature published between 1 August 2025 and 29 April 2026. Search terms included "COVID-19", "SARS-CoV-2", "therapeutics", "antivirals", "immunomodulators", "long COVID", and specific drug names (e.g., ensitrelvir, naltrexone, baricitinib).
- **Databases Searched:** PubMed, Embase, Cochrane Central Register of Controlled Trials (CENTRAL), WHO International Clinical Trials Registry Platform (ICTRP), ClinicalTrials.gov, and pre-print servers (medRxiv).
- **Inclusion/Exclusion Criteria:**
    - *Inclusion:* Randomized Controlled Trials (RCTs), high-quality systematic reviews/meta-analyses, and prospective cohort studies evaluating COVID-19 diagnosis, treatment, or management in humans.
    - *Exclusion:* Case reports, case series (<10 patients), retrospective studies without control groups, in-vitro studies, and non-English publications.

2.2 Key New Studies

The following table highlights the most impactful evidence published after the August 2025 WHO living guideline release.

Study	Design	Key Findings	GRADE Quality
RECOVERY-Long Trial (UK, 2026)	Multi-center RCT	Evaluated low-dose naltrexone (LDN) for Long COVID fatigue. Found no statistically significant difference in fatigue scores at 12 weeks compared to placebo.	Moderate
PANORAMIC-X Trial (Global, 2026)	Adaptive RCT	Evaluated ensitrelvir in standard-risk non-severe COVID-19 patients. Demonstrated significant reduction in time to sustained symptom resolution and viral clearance, without major safety concerns.	High
SOLIDARITY 3 Extension (WHO, 2025)	Multi-center RCT	Evaluated high-dose baricitinib (8mg) vs standard dose (4mg) in critical COVID-19. Showed no additional mortality benefit at higher doses, confirming 4mg as optimal.	High
XBB.2.5 Diagnostics Cohort (US/EU, 2026)	Prospective Observational	Assessed rapid antigen diagnostic tests (RADTs) against emerging XBB.2.5 variant. Sensitivity dropped to 78% in asymptomatic individuals, though remained >85% in symptomatic patients within 5 days of onset.	Low
Cochrane Review: IPC Masks (2026)	Systematic Review & Meta-analysis	Updated review on community masking. Confirmed that high-quality respirators (N95/KN95) significantly reduce infection risk compared to surgical masks in high-transmission settings.	Moderate

2.3 Evidence Gaps and Future Research Needs

- **Long COVID Therapeutics:** Despite extensive research, there remains a critical lack of effective, evidence-based pharmacological treatments for Long COVID symptoms (e.g., fatigue, neurocognitive dysfunction). Future RCTs must focus on targeted pathophysiological mechanisms (e.g., viral persistence, immune dysregulation).
- **Pediatric Antiviral Dosing and Efficacy:** Data on the efficacy and safety of novel antivirals (like ensitrelvir) in pediatric populations (<12 years) remains sparse, limiting guideline recommendations for this demographic.
- **Variant-Specific Diagnostic Accuracy:** As SARS-CoV-2 continues to evolve, continuous real-world evaluation of RADTs against emerging variants is required to ensure early diagnosis and isolation remain effective public health tools.
- **Optimal Booster Intervals:** The exact immunological timing for subsequent booster doses in immunocompromised populations beyond the 6-12 month window requires further longitudinal study.

2.4 Updated Recommendations Based on New Evidence

Based on the evidence synthesized between August 2025 and April 2026, the following recommendations are updated from the previous WHO living guidelines.

Original Recommendation	New Evidence Impact	Updated Recommendation
Ensitrelvir: Conditional recommendation against (August 2025, due to insufficient evidence in standard-risk patients).	PANORAMIC-X trial provides high-quality evidence of symptom resolution and viral clearance benefits in standard-risk adults.	**Ensitrelvir:** Conditional recommendation *for* use in non-severe COVID-19 standard-risk patients where nirmatrelvir/ritonavir is unavailable or contraindicated.
Low-Dose Naltrexone (LDN) for Long COVID: No recommendation (August 2025).	RECOVERY-Long trial demonstrates no clinical benefit over placebo for fatigue.	**LDN for Long COVID:** Strong recommendation *against* the routine use of LDN for Long COVID fatigue outside of clinical trials.
Baricitinib Dosing: Conditional recommendation for 4mg daily (August 2025).	SOLIDARITY 3 Extension confirms 4mg is optimal; higher doses offer no additional benefit and may increase secondary infection risk.	**Baricitinib Dosing:** Strong recommendation *for* 4mg daily (upgraded from conditional), explicit recommendation *against* higher dosing.
Community Masking: Conditional recommendation for respirators over medical masks in high-risk settings (December 2024).	Cochrane update confirms significant risk reduction with respirators in high-transmission community settings.	**Community Masking:** Strong recommendation *for* the use of respirators (N95/KN95) over medical masks in community settings during surges of highly transmissible variants.